

SUMON SOURABH GHOSH

SDET – Backend & DevOps Focus

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PROFESSIONAL SUMMARY

Backend-focused SDET with 2 years of experience specializing in API testing, backend validation, unit testing, and integration testing across distributed microservices. Proficient in Python and Node.js with strong hands-on experience in CI/CD pipeline tools (GitHub Actions, Jenkins), automation frameworks (pytest, requests), and cloud computing on AWS (EC2, S3, CloudWatch). Skilled in Docker-based containerization, container orchestration, Infrastructure as Code with Terraform, and performance optimization using k6. Experienced in TDD/BDD methodologies, security-aware API validation, Agile/Scrum workflows, and monitoring tools including CloudWatch. Proven ability to drive reliability through Blue/Green deployment strategies and continuous delivery.

WORK EXPERIENCE

Tata Consultancy Services — Assistant System Engineer

Jun 2022 – Feb 2024

- Developed and maintained backend services using Node.js and REST APIs; implemented comprehensive API testing, request/response analysis, and end-to-end integration testing using the requests automation framework, incorporating security-aware validation for common vulnerabilities (auth bypass, injection, improper error handling).
- Designed and authored 300+ unit and integration test cases (functional, regression, exploratory, and BDD-style using Gherkin) with pytest across distributed microservices, ensuring contract adherence and inter-service communication reliability.
- Configured and maintained CI/CD pipeline tools using GitHub Actions and Jenkins, automating build, test, and deployment stages with Blue/Green deployment strategies, reducing deployment lead time by 30%.
- Applied data-consistency validation scripts across distributed systems, identifying 30+ edge-case failures and reducing data-integrity defects by 25%.
- Provisioned and managed cloud computing infrastructure on AWS (EC2, S3, IAM) using Terraform (Infrastructure as Code), reducing manual environment setup time by 40% and ensuring cross-environment consistency.
- Leveraged monitoring tools (CloudWatch) for production incident resolution and root-cause analysis, cutting mean time-to-resolution by 20% and reducing critical post-deployment bugs by 15%.
- Contributed to backlog refinement, story estimation, and sprint retrospectives within Agile/Scrum; mentored 3 junior engineers on test automation best practices and CI/CD workflows.

PROJECTS

applier.cloud – AI Resume Builder SaaS

Jan 2025 – Present | applier.cloud | github.com/sghoshm/applier.cloud

- Architected and deployed a full-stack AI Resume Builder SaaS using the MERN stack (MongoDB Atlas, Express, React, Node.js) with Groq LLM API (Llama 3.3 70B) for resume tailoring and cover letter generation.
- Implemented JWT + Google OAuth 2.0 authentication with bcrypt password hashing, refresh token rotation, and Passport.js; secured API routes with role-based plan middleware (Free/Pro/Elite).
- Built PDF parsing pipeline using pdf2json with AI-assisted structured extraction and PDF export using @react-pdf/renderer with dynamic resume templates.
- Designed ATS scoring system with keyword alignment analysis returning match scores, missing keywords, and improvement suggestions via LLM; applied performance optimization to reduce response latency.
- Deployed backend on Render and frontend on Netlify with CI/CD pipeline automation via GitHub; configured MongoDB Atlas M0 free cluster — full infrastructure at \$0/month.
- Designed freemium SaaS pricing model (Free/Pro/Elite) with AI credit system, plan-gating middleware, and Lemon Squeezy payment integration scaffold.

OpenEarth – Data Aggregation & Visualization Platform

Jan 2025 – Present | earthint.netlify.app | github.com/sghoshm/open-earth

- Developed a full-stack platform aggregating Earth-related datasets via multiple external APIs; implemented unit and integration testing using automated pytest scripts for accuracy, consistency, and security validation.
- Engineered backend data-retrieval and processing logic; implemented interactive frontend visualization with performance optimization for efficient data rendering.
- Deployed via CI/CD pipeline tools on Netlify with health monitoring; evaluated system behaviour under 15+ diverse input and edge-case scenarios.

Nexus OSINT Platform

Jan 2025 – Present | [nexusint.netlify.app](#) | [github.com/sghoshm/nexus-osint](#)

- Constructed backend system aggregating real-time public data from multiple APIs; containerized with Docker and managed container orchestration and deployments via GitHub Actions CI/CD pipelines.
- Designed modular microservices architecture; optimized data-pipeline performance and applied structured integration testing across all data sources.

OpenJarvis – AI Assistant with API Integration & Automation

Jan 2025 – Present | [github.com/sghoshm/open-jarvis](#)

- Built a Python AI assistant integrating external APIs, OS-level automation, and dynamic command workflows with safety checks.
- Completed end-to-end integration testing and unit testing across 20+ input scenarios following TDD principles, covering command-parsing, execution, and response flows.

Personal Project: Self-Hosted Video Streaming Platform

Jun 2020 – Feb 2024

- Designed and deployed full-lifecycle backend (auth, session management, media delivery) on Linux with SSL and Cloudflare tunnelling; managed DNS routing and secure traffic configuration.
- Owned full development, deployment, testing, and monitoring pipeline; performed load and performance optimization analysis under concurrent traffic using k6.

Hand Gesture Controlled Mouse (Python)

Mar 2023 | [github.com/sghoshm/hand-gesture-mouse](#)

- Developed real-time computer-vision gesture-recognition system; applied performance optimization on frame-processing and validated edge-case inputs systematically.

EDUCATION

Bengal Institute of Technology and Management

Jun 2024 – Present

M.E / M.Tech – Electronics & Communication Engineering

Sathyabama Institute of Science and Technology

Aug 2018 – Sep 2022

B.E / B.Tech – Electronics & Communication Engineering

LICENSES & CERTIFICATIONS

Python

freeCodeCamp | Issued May 2026 | Credential ID: aimingup-pyv9 | [View Certificate](#)

Assistant System Engineer

micro1 | Issued Apr 2026 | [View Certificate](#)

Azure: Create a Virtual Machine and Deploy a Web Server

Coursera | Issued Apr 2026 | Credential ID: QPK11BGJOHP7 | [View Certificate](#)

TECHNICAL SKILLS

Testing & QA: API Testing (Postman), Unit Testing, Integration Testing, Functional / Regression / Exploratory Testing, Test Case Design, BDD / TDD Principles, Gherkin-style Test Design, Security-Aware API Validation, Performance Testing (k6), Data & Edge-Case Testing, Defect Lifecycle Management

Automation Frameworks: pytest, requests, API automation scripting, Selenium (familiar), JUnit (familiar)

Programming: Python, JavaScript (Node.js)

Backend & APIs: REST APIs, JWT Authentication, API Validation, Microservices (contract & inter-service testing), Node.js

CI/CD Pipeline Tools: GitHub Actions, Jenkins, Blue/Green & Canary Deployment Strategies, Automated Build/Test/Deploy Pipelines

Cloud Computing & Infra: AWS (EC2, S3, CloudWatch, IAM, Lambda, RDS), Docker, Container Orchestration, Infrastructure as Code – Terraform, SSL/HTTPS, Cloudflare DNS & Tunnelling, Linux Server Administration

Monitoring Tools: CloudWatch, k6, Health Monitoring, Log Analysis, Root-Cause Analysis

Databases: MongoDB, Firebase, PostgreSQL / SQL (queries, data extraction & validation)

Tools & Practices: Git, GitHub, Postman, Agile/Scrum, Performance Optimization, TDD, BDD